INTELLIMINE

IoT Smart Gas Monitoring System

خوذة ذكية لعمال المناجم للكشف عن الغازات السامة

POWERED BY ESP32 MICROCONTROLLER

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PROBLEM STATEMENT

The Silent Threat In Mining

Fire & Explosion Hazards
Uncontrolled gas leaks create immediate risk of ignition.

Severe Health Risks
Toxic gas inhalation threatens lives of underground personnel.

Environmental Damage
Spreading toxins compromise sensitive industrial ecosystems.

CRITICAL MANDATE

Continuous monitoring is essential to ensure safety.

PROJECT OBJECTIVE

Empowering Safety Through IoT

01
Environmental Sensing
Continuous monitoring of ambient parameters to safeguard mining zones.

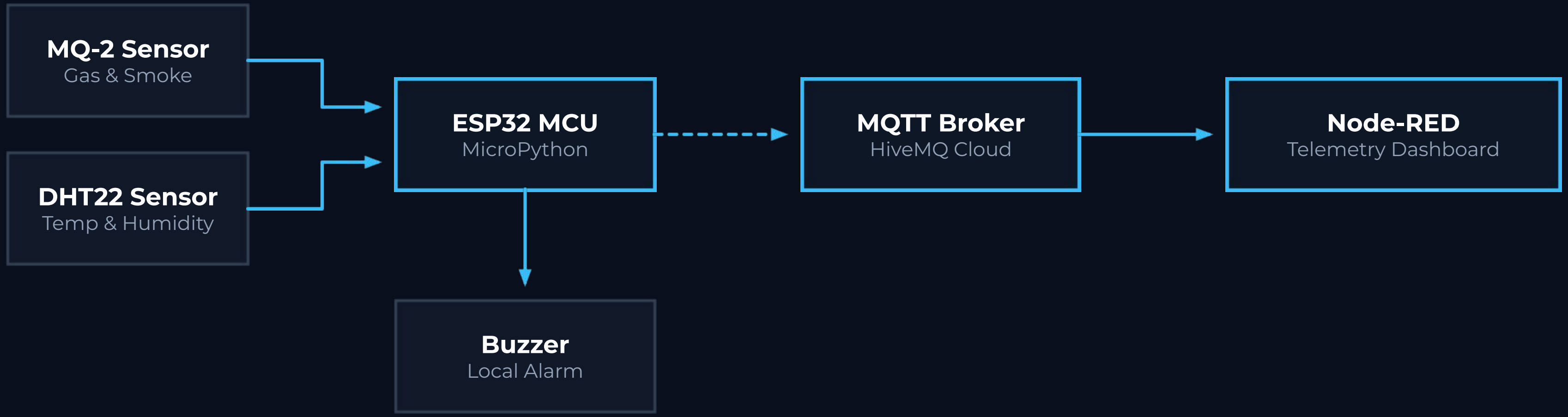
02
Instant Alerting
Immediate warning dispatch when hazardous conditions are detected.

CORE MISSION OBJECTIVES

- **Monitor Gas Concentration:** Protect personnel from toxic leaks.
- **Measure Temperature & Humidity:** Track structural atmospheric trends.
- **Dangerous Gas Alerts:** Trigger real-time, life-saving smart alarms.
- **Real-Time Web Dashboard:** Centralized visual data telemetry.

SYSTEM ARCHITECTURE

System Data Flow



SYSTEM WORKFLOW

How It Works

01
Sensor Reading

ESP32 micro-controller reads raw analog and digital values from sensors.

02
Data Conversion

Raw atmospheric readings are packaged into a structured JSON payload.

03
MQTT Publish

ESP32 publishes JSON data securely to HiveMQ Cloud via MQTT.

04
Node-RED Subscribe

Central Node-RED system subscribes to the active MQTT telemetry topic.

05
Dashboard Render

Web interface visualizes structural data streams and metrics in real-time.

06
Smart Alarm

Buzzer is triggered immediately if gas levels exceed safe safety thresholds.

REAL-TIME TELEMETRY CYCLE

This complete data loop ensures sub-second latency from toxic gas detection to local and remote hazard alerting.

NODE-RED TELEMETRY

Real-Time Monitoring

Live atmospheric feeds and safety status updated with sub-second latency.

Temperature
24.5 °C

Humidity
58.2 %

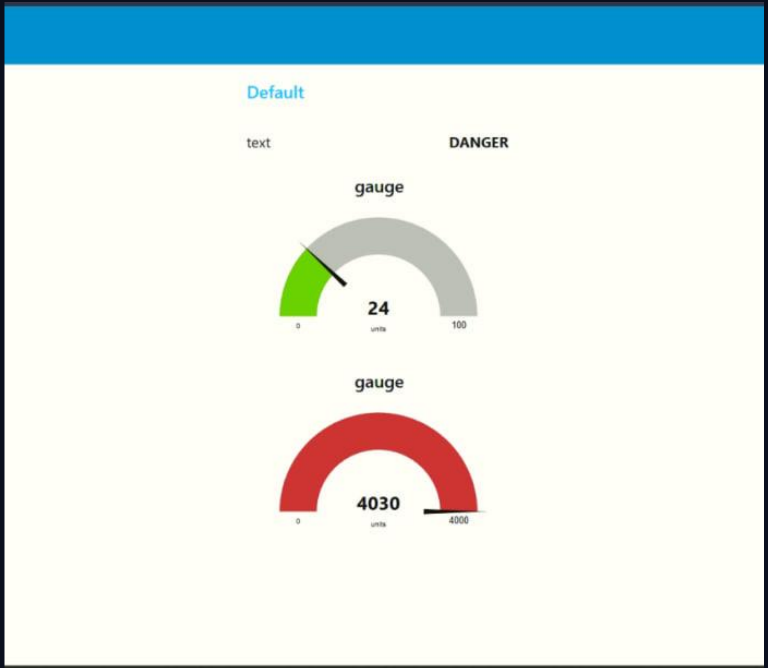
Gas Level
124 ppm

System Status
SAFE

NODE-RED INTEGRATION

Real-Time Monitoring using Node-RED Dashboard

Aggregates environmental parameters from deep-mine sensor units and delivers diagnostic visualization with dynamic gauges.



SYSTEM SIMULATION

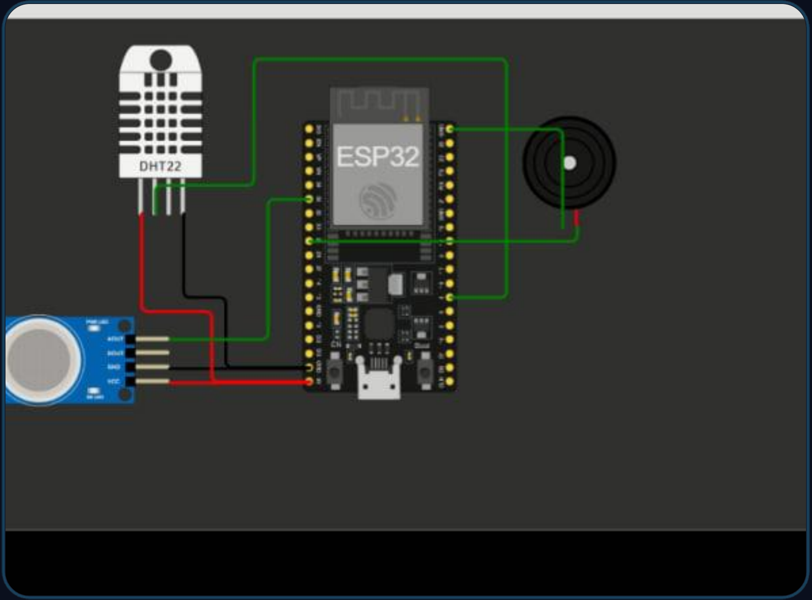
Simulation of Wokwi

01 ESP32 MCU
Core microcontroller executing control logic and network stack.

02 MQ-2 Sensor
Gas and smoke sensor simulation for hazard detection.

03 DHT22 Sensor
High-precision temperature & humidity simulation.

04 Buzzer Alarm
Simulated audio buzzer for instant local hazard notification.



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Thank You

Questions?

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